DOCETAXEL

Fluid retention — steroid premed

Neutropenia

CHEMICAL NAME Semi-synthetic taxane (C43H53NO14, trihydrate) from European yew (*Taxus baccata*)

CLASS Taxane (microtubule-stabilising) cytotoxic

TRADE NAMES Taxotere · Docefrez · Docetaxel injection

THERAPEUTIC USES / INDICATIONS

- Breast cancer (adjuvant TAC/TC, metastatic, with trastuzumab).
- Prostate cancer (metastatic, with prednisolone/ADT).
- Non-small cell lung cancer; gastric and head and neck cancers.

ADMINISTRATION

- **IV** infusion over 1 h.
- Premedication: oral dexamethasone 8 mg twice daily for 3 days starting the day before (fluid retention, hypersensitivity); 3 doses for prostate cancer.
- Contains ethanol in some products (driving, alcohol intolerance).

DOSAGES

- **Breast:** 75–100 mg/m² every 3 weeks.
- **Prostate:** 75 mg/m² every 3 weeks with prednisolone 5 mg twice daily.
- **NSCLC:** 75 mg/m².
- Avoid if bilirubin > ULN or ALT/AST >1.5× ULN with ALP >2.5× ULN.

PHARMACOKINETICS

- Metabolised by CYP3A4 in liver.
- Excreted in faeces. **t½:** ~11 h (terminal).

PHARMACODYNAMICS

- Promotes **tubulin assembly** and inhibits depolymerisation → mitotic arrest → apoptosis (more potent than paclitaxel).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Alopecia, fatigue.
- Nail changes (onycholysis), mucositis.
- Diarrhoea, nausea, taste change.

ADVERSE EFFECTS (SERIOUS)

- Febrile neutropenia (dose-limiting).
- Fluid retention (oedema, pleural effusion — cumulative).
- Hypersensitivity; peripheral neuropathy.
- Cystoid macular oedema (epiphora, blurred vision); severe skin reactions; enterocolitis.

CONTRAINDICATIONS & CAUTIONS

- Neutrophils <1.5 × 10^9/L.
- Severe hepatic impairment.
- Severe hypersensitivity (polysorbate 80).
- Pregnancy, breastfeeding.

KEY INTERACTIONS

- **Strong CYP3A4 inhibitors (ketoconazole, clarithromycin, ritonavir):** ↑ toxicity; avoid.
- **Live vaccines:** contraindicated.

NURSING MANAGEMENT

- Confirm dexamethasone premedication taken.
- FBC, LFTs before each cycle; G-CSF if prescribed.
- Monitor weight and oedema.
- Monitor for hypersensitivity during first infusions.
- Nail care (frozen gloves/socks may reduce nail toxicity).

PATIENT EDUCATION

- Take dexamethasone tablets as instructed before and after treatment.
- Report fever ≥38 °C urgently (neutropenia peaks days 5–8).
- Report swelling, weight gain, breathlessness, watery eyes or blurred vision.
- Keep nails short and clean.

KEY POINTS

- Taxane; breast, prostate, lung cancer.
- Febrile neutropenia, fluid retention.
- Dexamethasone premed 3 days.
- CYP3A4 interactions; hepatic impairment.